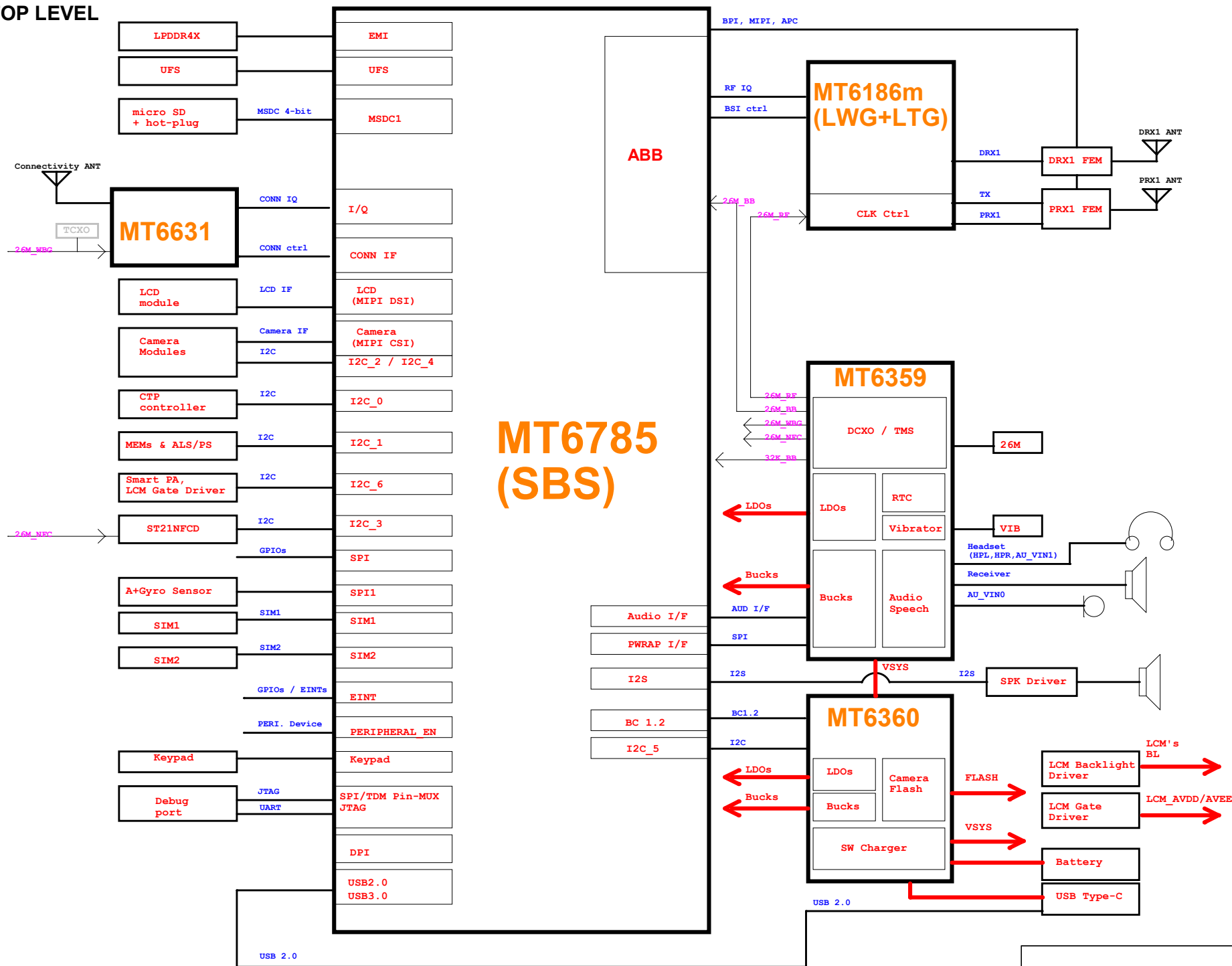
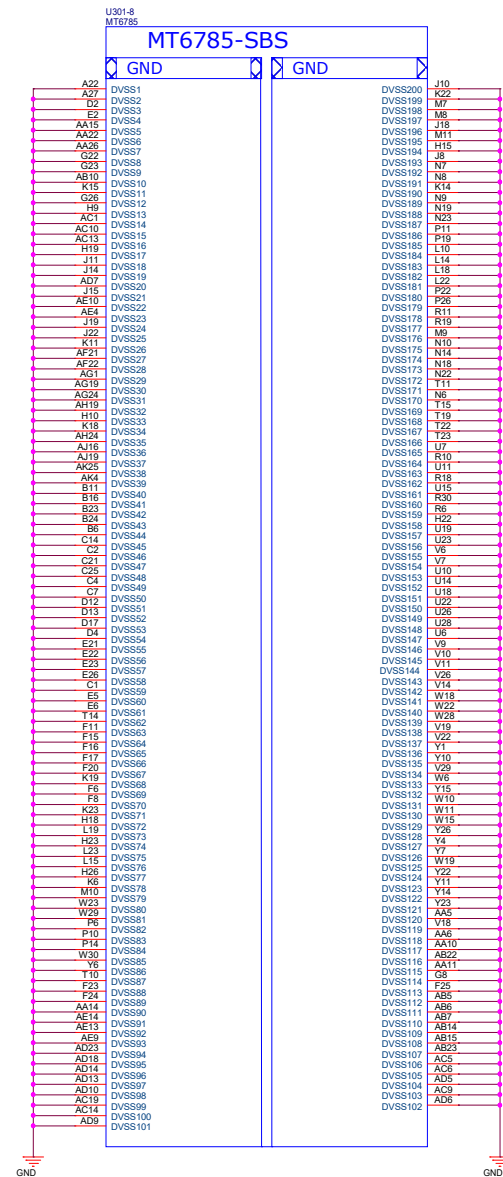
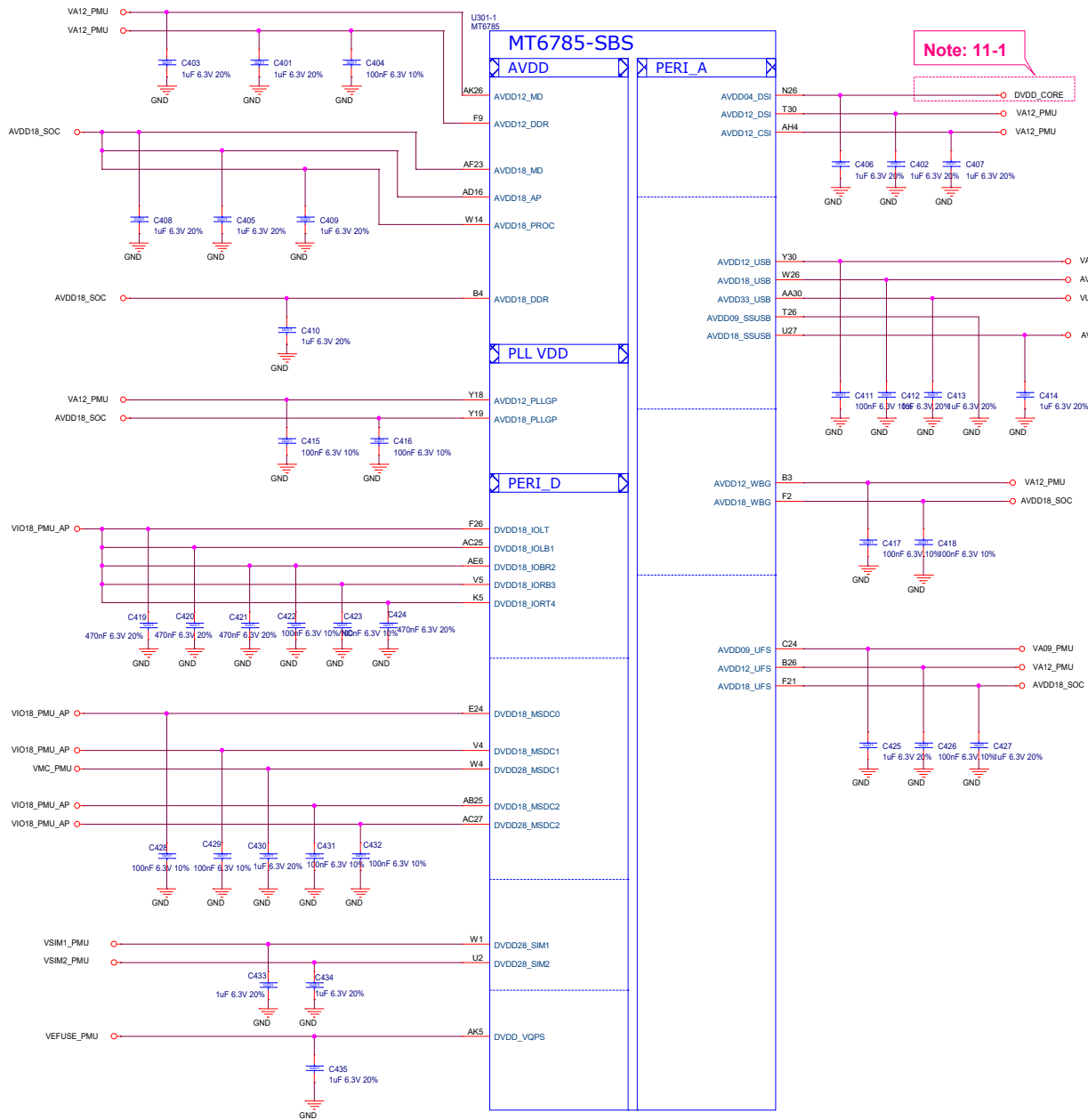


Project : MT6785 LPDDR4X
REF SCH TOP LEVEL

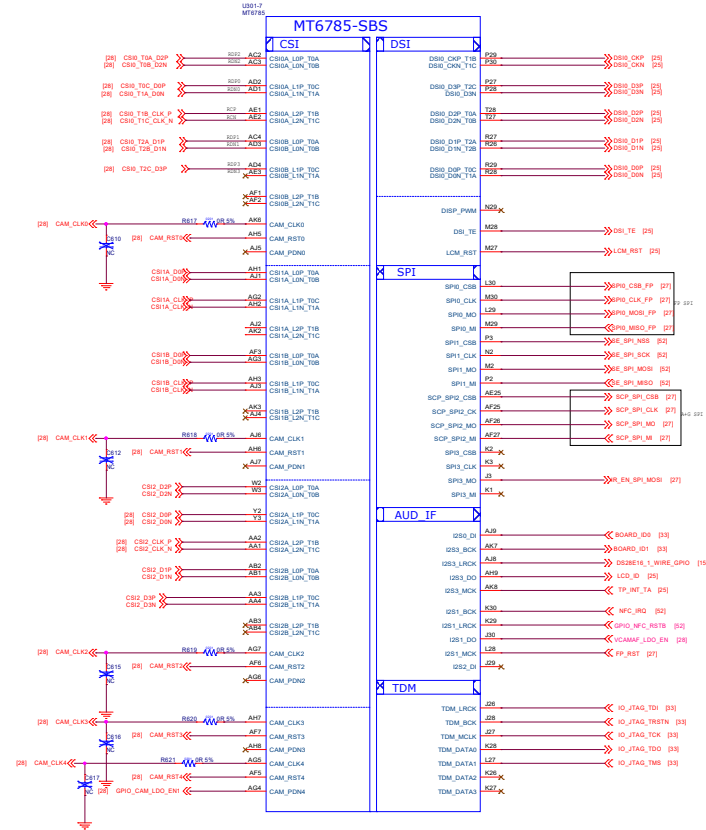
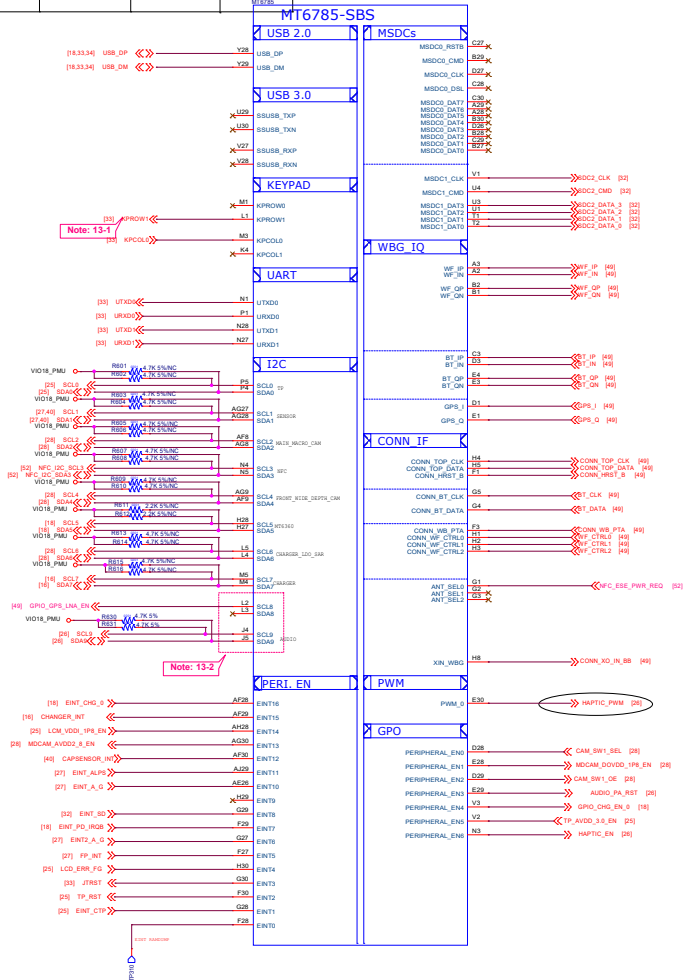




Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: For D-PHY application, connect MIPI Power domain AVDD04_DSI to DVDD_CORE.
For C-PHY application, connect MIPI Power domain AVDD04_DSI to VA12_PMU.

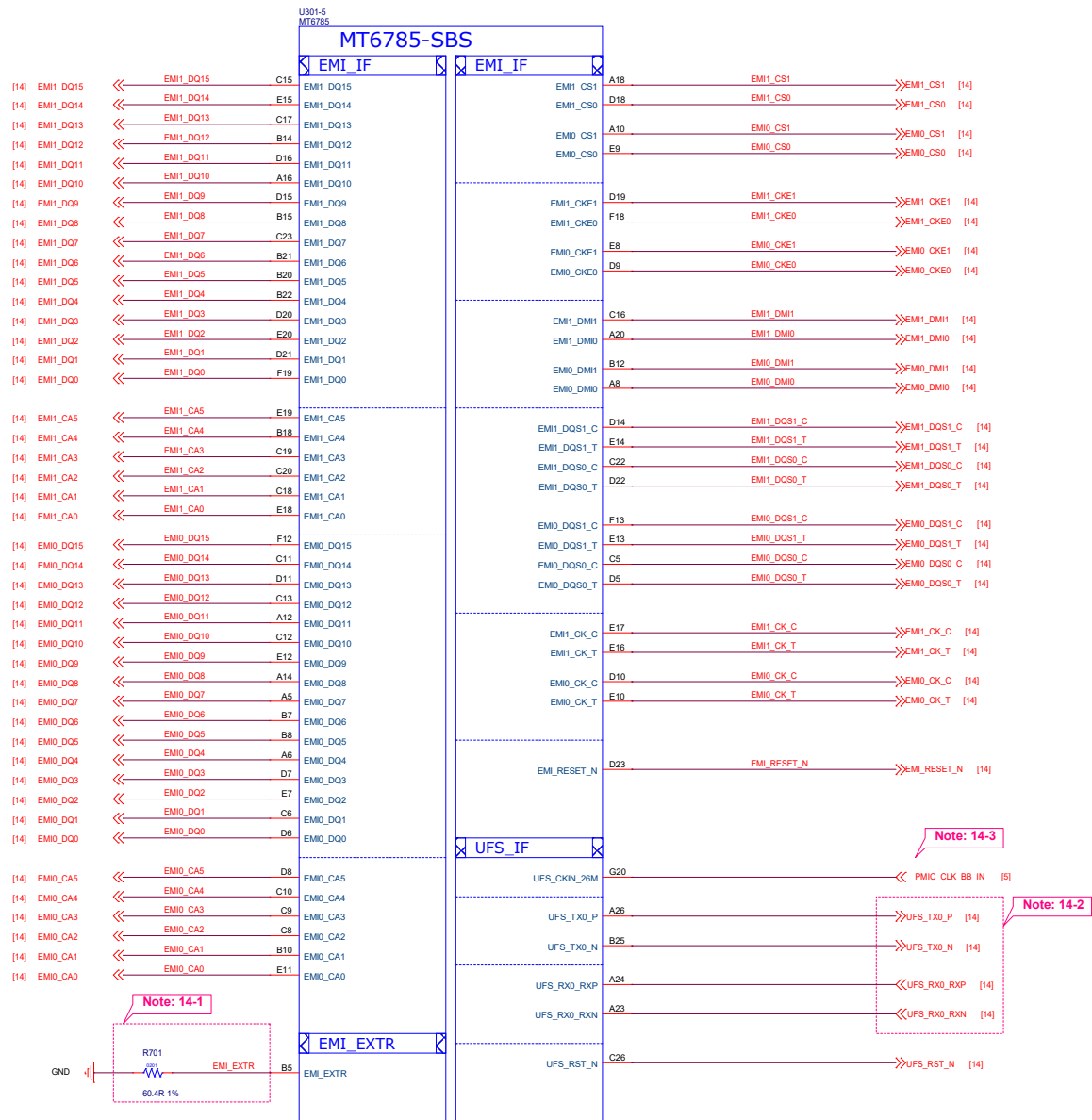
I2C	Device1	Device2	Device3	Device4	Device5	Device6
0	CTP					
1	TSL2540	BU27030NUC	AK09918C	Sar_SENSOR		
2	Main_CAM	Macro_CAM				
3	NFC ST21					
4	Front_CAM	Depth_CAM				
5	MT6360					
6	WL2865D-10/TR	Wide_CAM				
7	SC8551					
9	AW87559FCR	AW87389FCR				



Schematic design notice of "06_BB_2" page.

Note 13-1: Reserved for fast META mode detection

Note 13-2: For (I2C-8, I2C-9), must have pull up resistor 4.7K ohm for standard mode and fast mode; 2.2K ohm for fast mode plus and high speed mode.

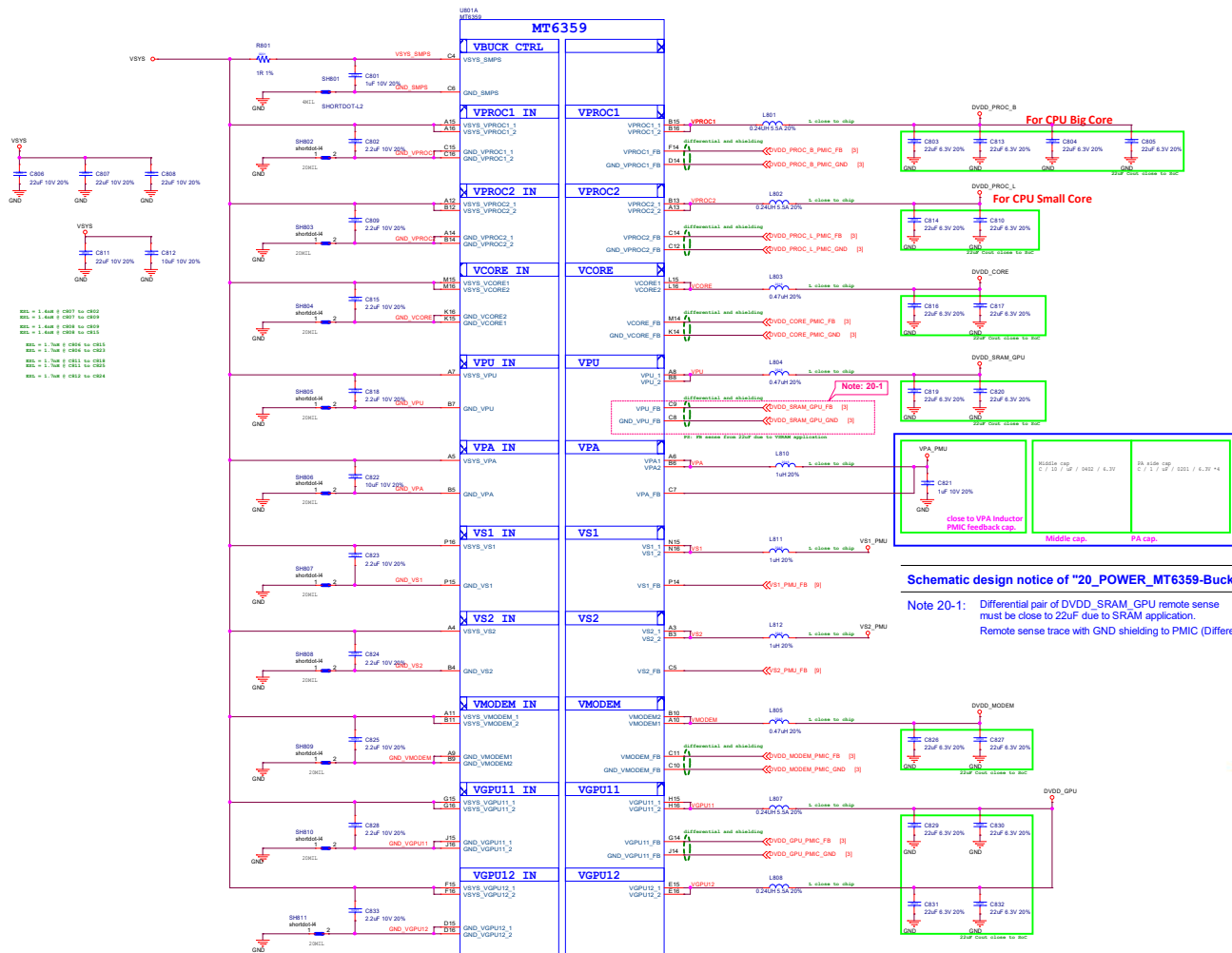


Schematic design notice of "14_BB_3_Interface" page:

Note 14-1: R4001 please select 60.4 ohm (1%) resistor

Note 14-2: Make sure TX/RX connection (T to in, R to out and P to t, N to c)

Note 14-3: UFS IP use 26MHz clock from PMIC XO_SOC (shared with ABB IF) and UFS device from PMIC XO_EXT.
All of their layout must be shielded by GND.



2. Although small size and high efficiency are major concerns, the inductor should have low core losses and low DCR (copper wire resistance). Efficiency data in MT6359 datasheet is based on 30mΩ DCR.

BUCK	Inductance (uH)	I _{OUT} MAX (A)	I _{TEMP} MAX (A)
VS1	1.0	2.6	2.0
VS2	1.0	2.6	2.0
VPA	1.0	4.0	3.4
VMODEM	0.47	4.8	4.2
VPROC1	0.24	6.3	5.0
VPROC2	0.24	6.1	4.8
VCORE	0.47	5.5	4.8
VGPU11	0.24	6.3	5.0
VGPU12	0.24	6.3	5.0
VPU (VSRAM_GPU)	0.47	1.3	0.6

Note: I_{OUT} Depends on inductance saturation. (-30% reduction from initial L value)

U801B
MT6359

MT6359

LDO IN

LDO

ALDO

DLDO

SLDO1

SLDO2

VREF

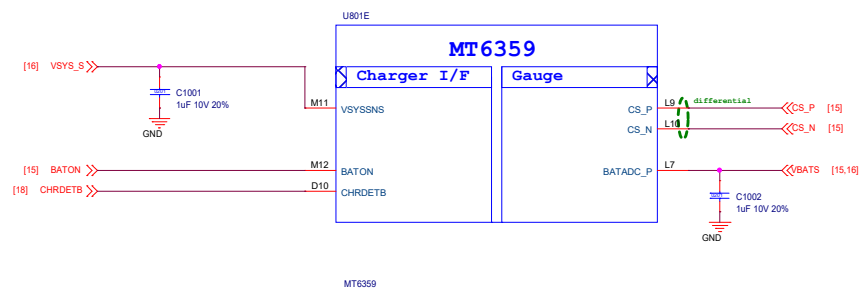
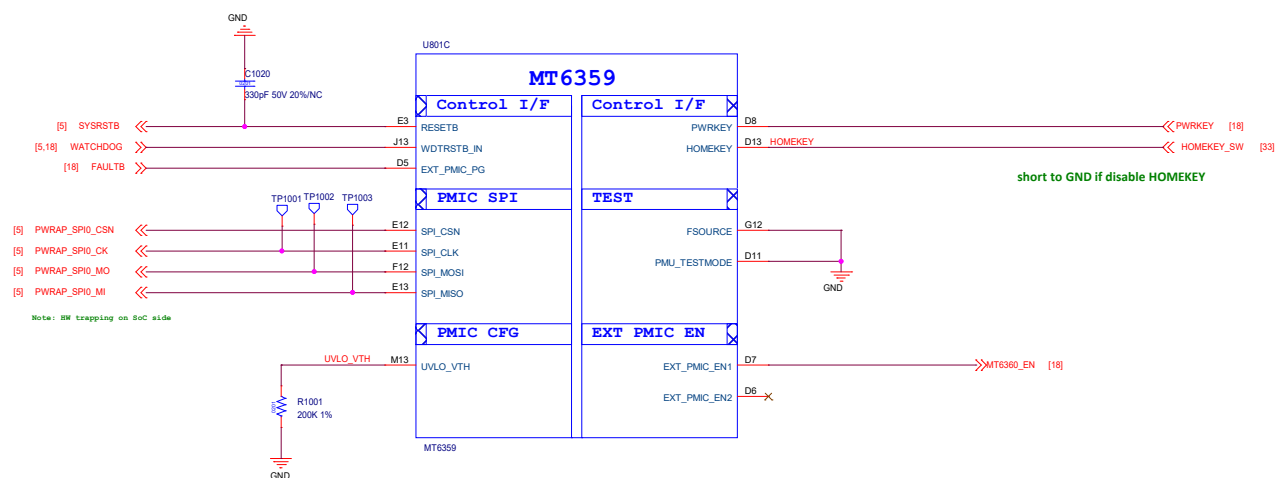
DIG Power

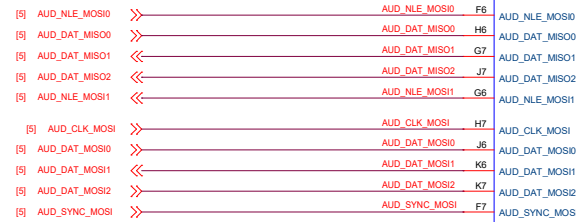
Close to PMIC

Note: For MT6631 need to configure the VCN33_2_PMU to 2.8V by PMIC SW.

L1 LDO Input Cap close to chip

<Core Design>



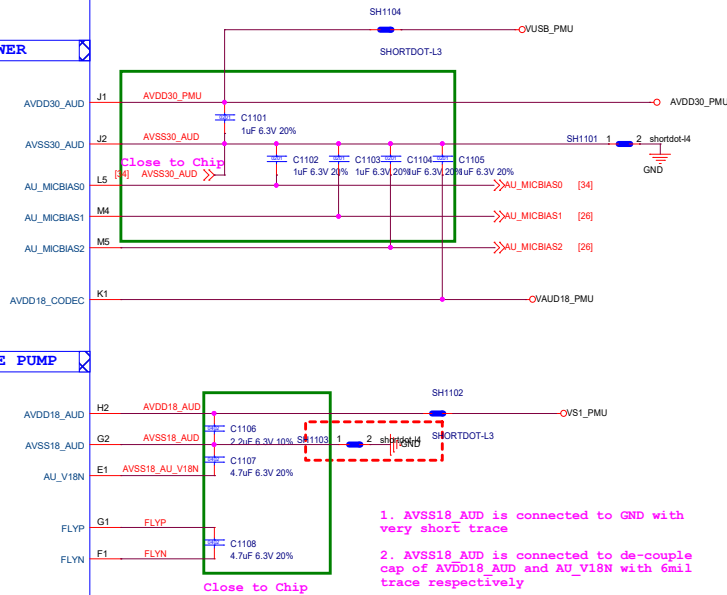
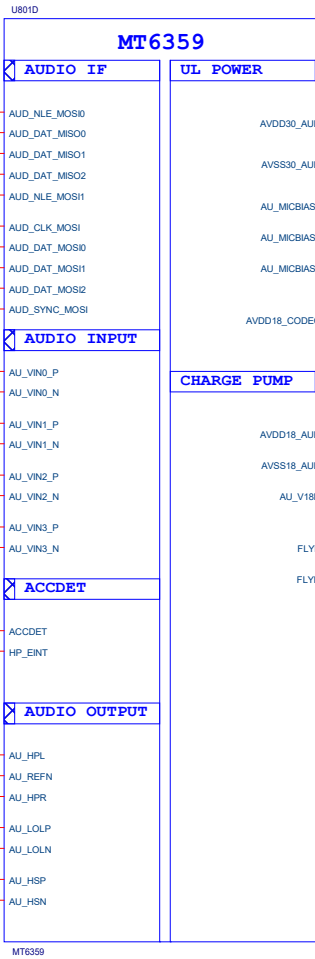
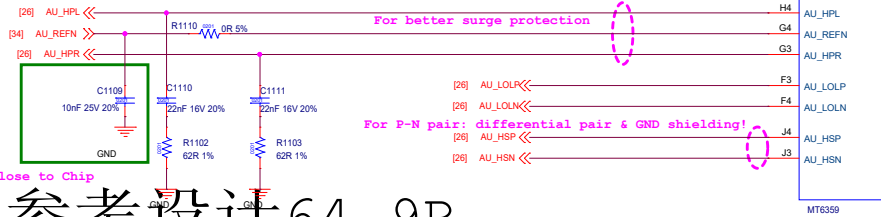


For better surge protection

For P-N pair: differential pair & GND shielding!

-AU_HPL and AU_HPR should be routed as single end signal and be guarded by GND, up and down, left and right respectively

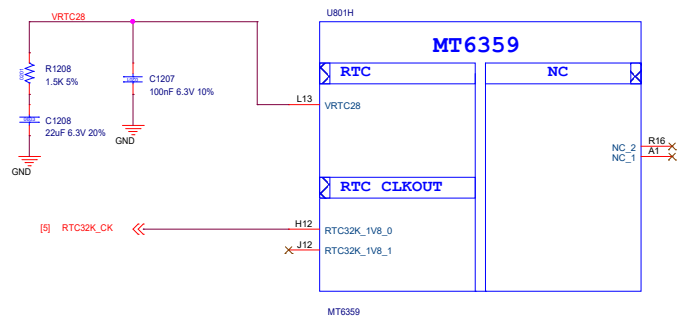
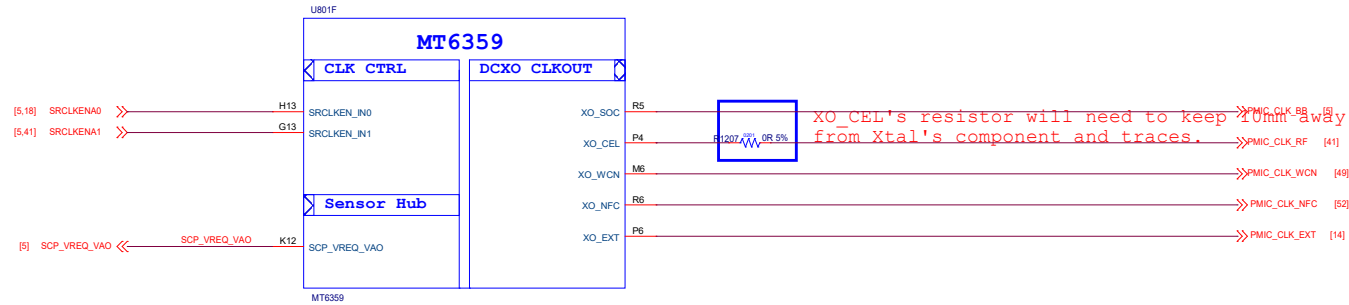
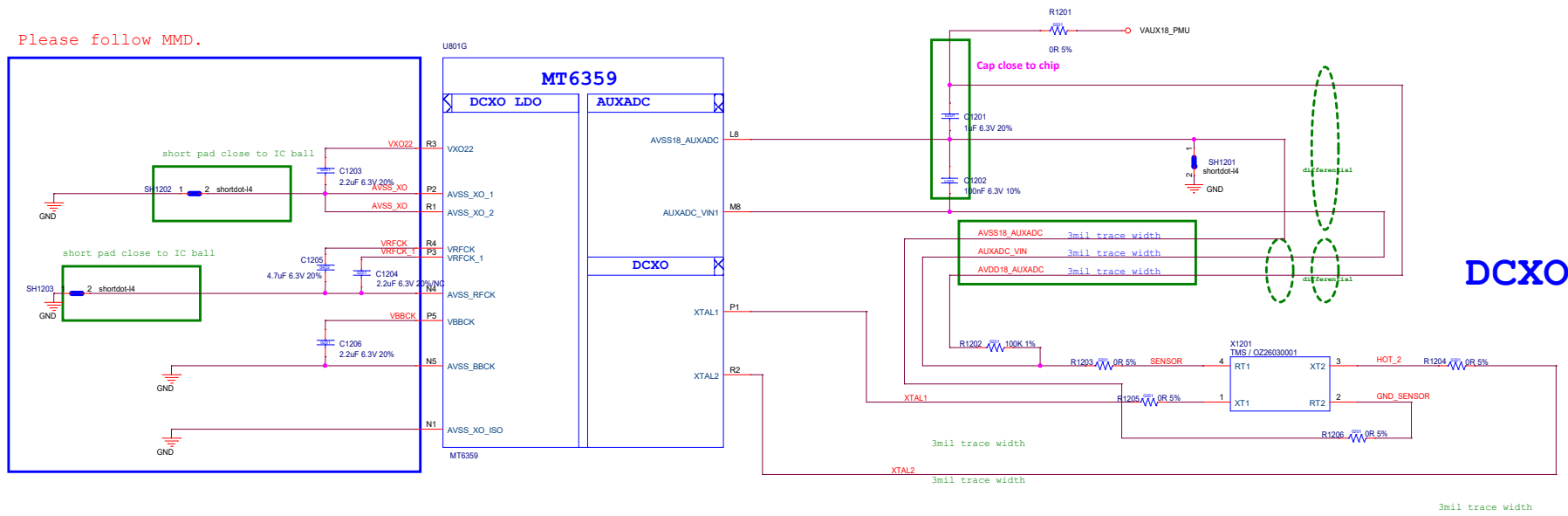
-The suggested layout pattern of AU_HPL/ AU_HPR/ AU_REFN is " GND AU_HPL AU_REFN AU_HPR GND"

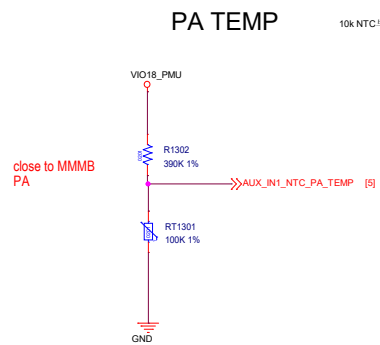


参考设计 64.9R

NOTICE

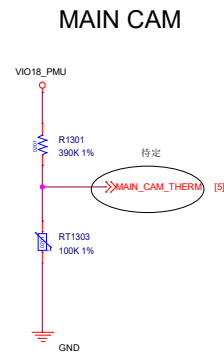
Please follow MMD.





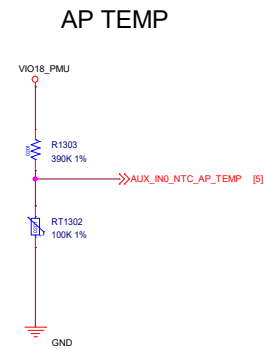
Thermistor to sense RF PA temperature

1. RT1301 must close to LTE Band 7 PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.



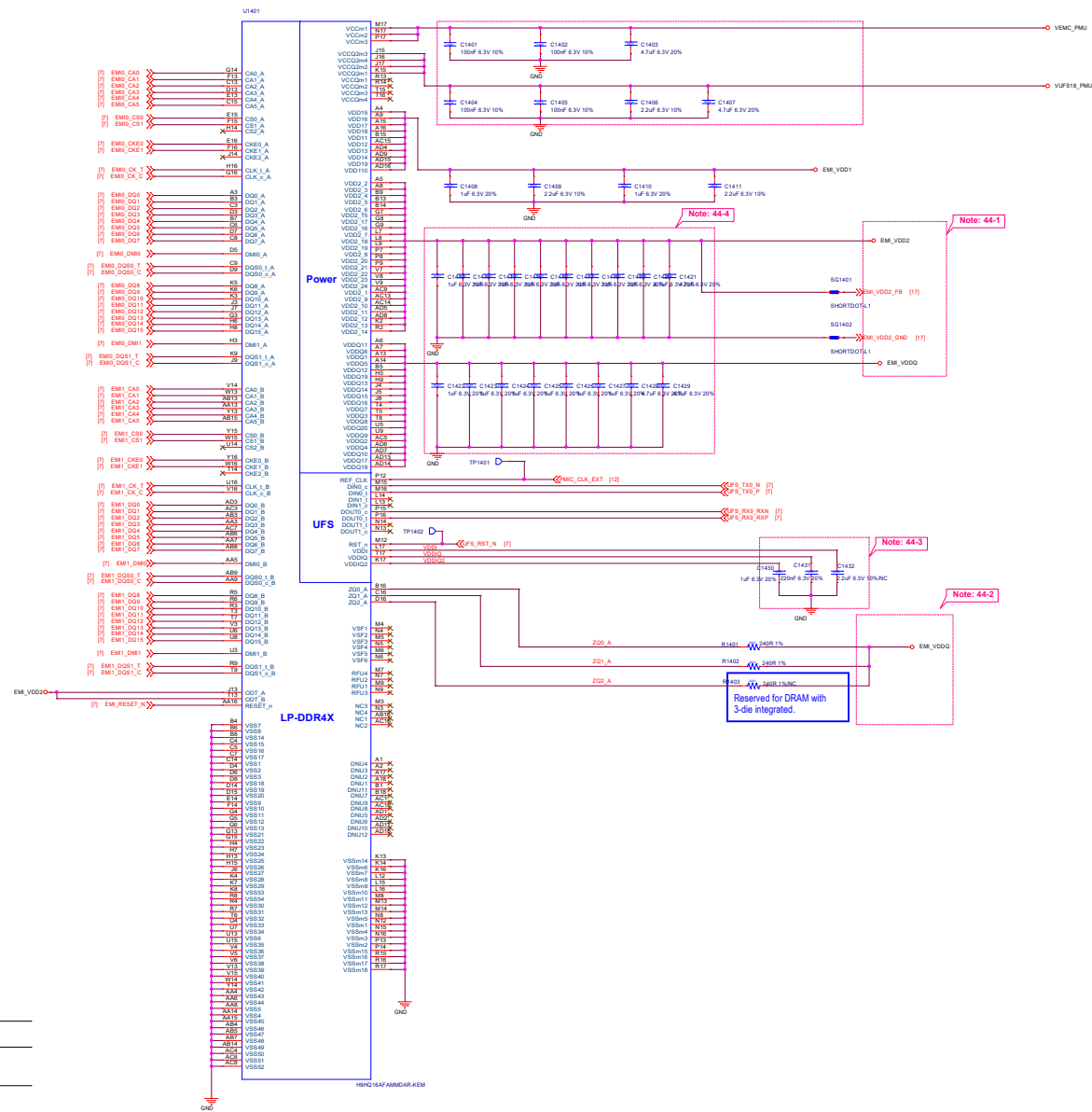
Thermistor to sense main cam temperature

1. RT1303 must close to main cam .
2. The distance is the shortest distance from package edge to edge.



Thermistor to sense AP temperature

1. RT1302 must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.



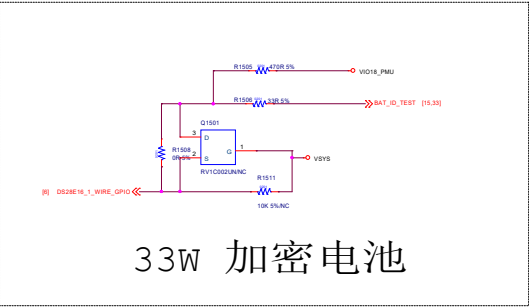
Schematic design notice of "44_Memory_UFS_LPDDR4X" page.

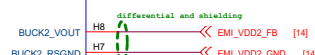
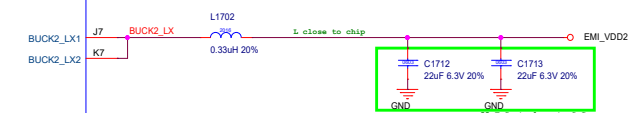
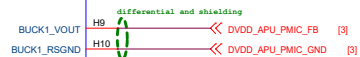
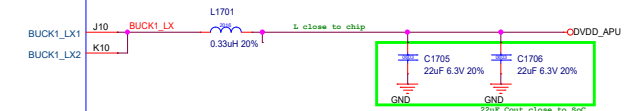
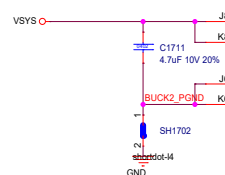
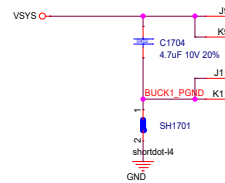
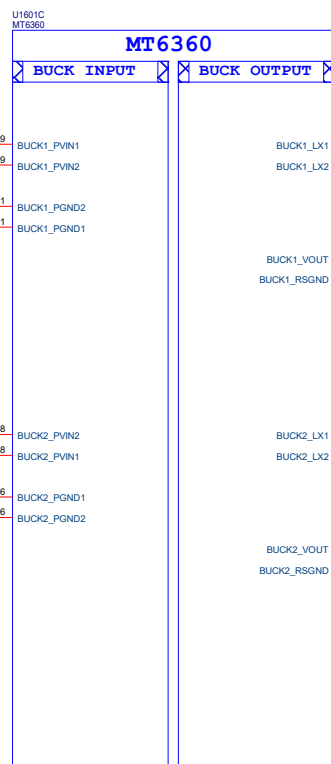
Note 44-1: Please refer to power supply related page select LDO7_VOUT / BUCK2_LX output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.

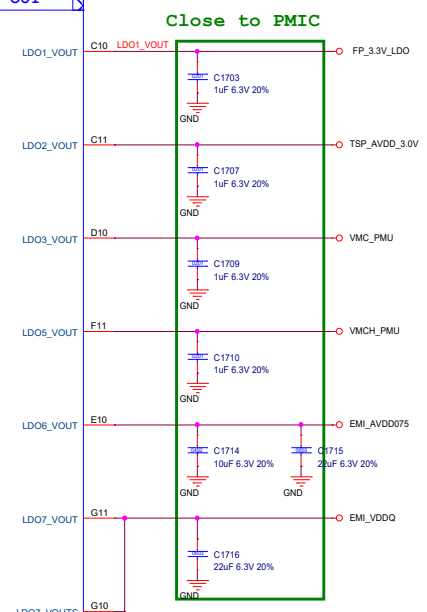
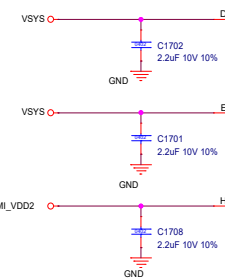
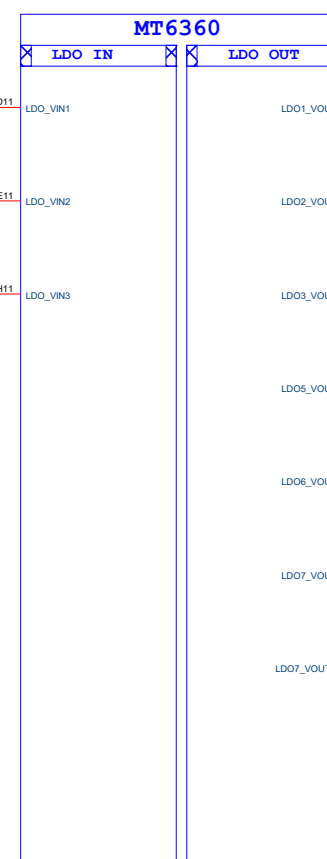
Note 44-3: Please refer to uMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of UFS.

Note 44-4: VDD2 VDDQ decoupling cap. closed to DRAM ball.
For other cap for PMIC (>10uF, at PMIC page),
please also refer to MMD and layout guide for placement.

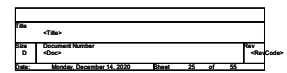
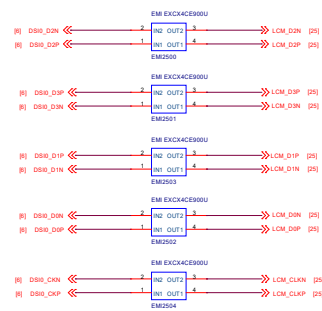




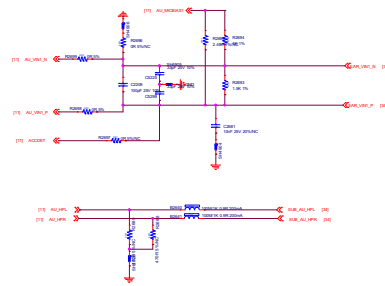
U1601D
MT6360



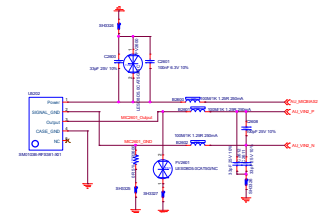
LDO1	VFP,	150mA
LDO2	VTP,	200mA
LDO3	VMC,	200mA
LDO5	VMCH,	800mA
LDO6	VMDDR,	300mA
LDO7	VDRAM2,	600mA



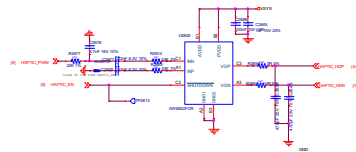
Earphone Audio



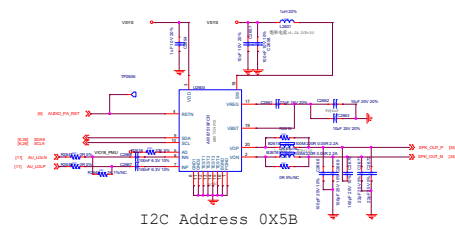
Sub MIC



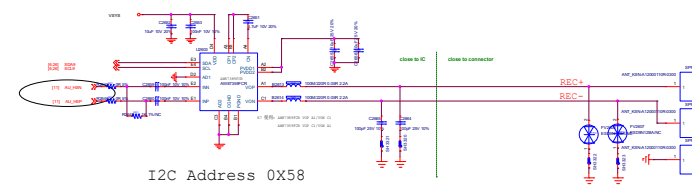
HAPTIC



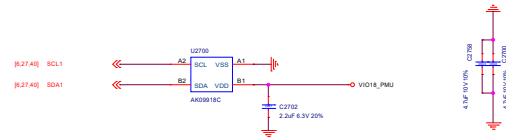
PA FOR SPK



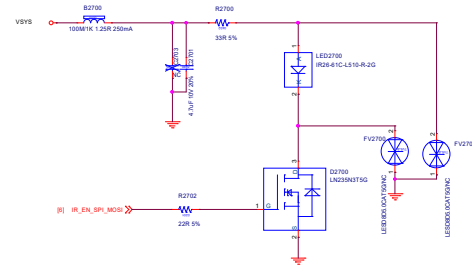
PA FOR REC



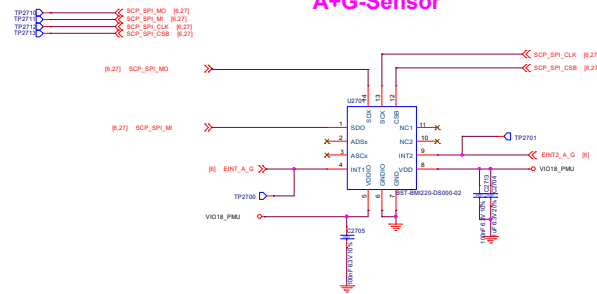
M SENSOR



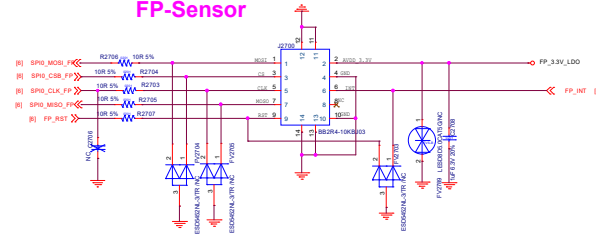
IRLED



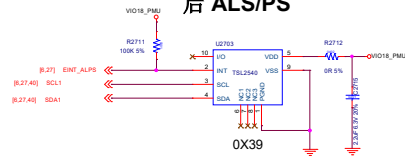
A+G-Sensor



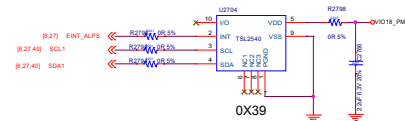
FP-Sensor



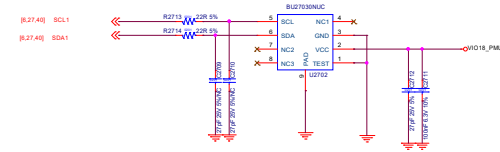
后 ALS/PS



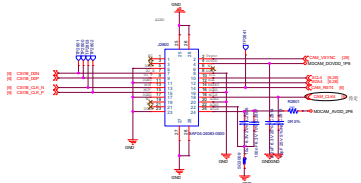
后 ALS/PS



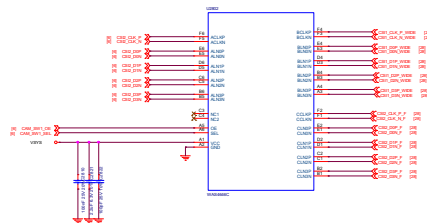
前面 ALS/PS



Depth Camera 2M

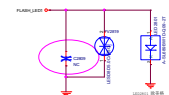


MIPI SW1

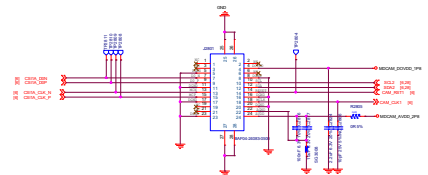


Ultra
Front

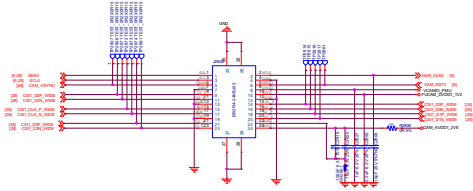
FLASH LED



Micro Camera 5M

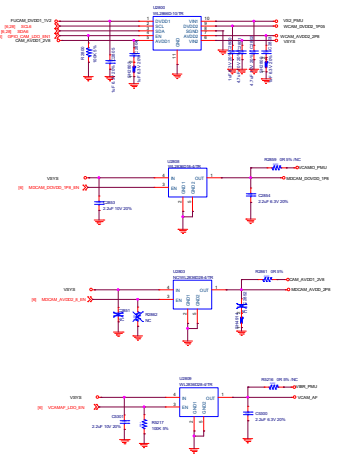


WIDE Camera 8M

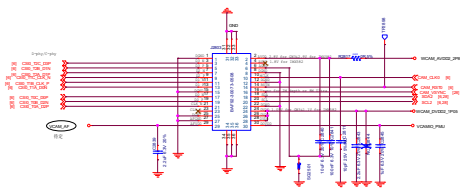


POWER LDO

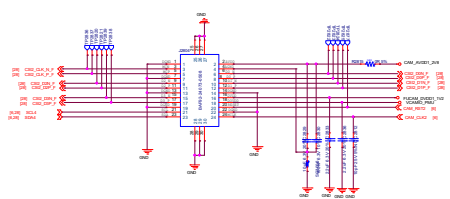
I2C Address 0X40



Main Camera 64M

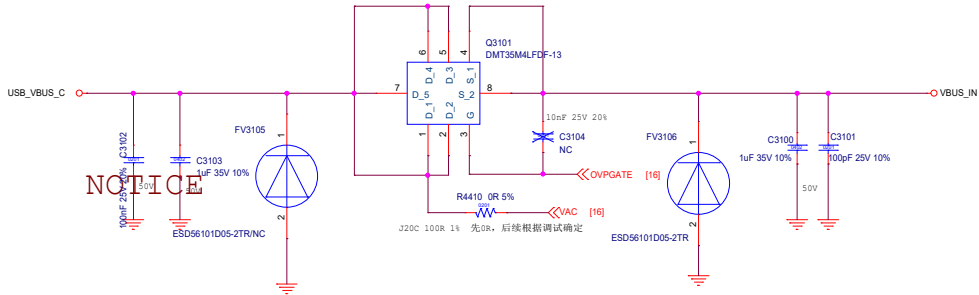


Front Camera 13M/16M

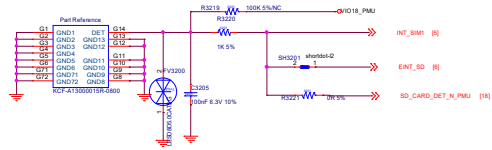
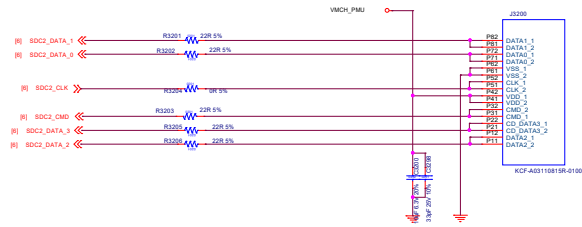


	AVDD	DVDD	IOVDD	APVDD	DC	MDLX	PEN	RESET
F_CAM	CAM_AVDD1_2M	VCAM_DVDD1_1V0	VCAM_IO_PBU		4	MDLX2		CAM_RST_N
B_CAM	VCAM_AVDD2_5M	VCAM_DVDD2_1V0	VCAM_IO_PBU		2	MDLX0		CAM_RST_N
M_CAM	CAM_AVDD1_5M	VCAM_DVDD1_1V0	VCAM_IO_PBU		2	MDLX1		CAM_RST_N
L_CAM	CAM_AVDD1_2M	VCAM_DVDD1_1V0	VCAM_IO_PBU		4	MDLX0		CAM_RST_N
W_CAM	CAM_AVDD1_8M	VCAM_DVDD1_1V0	VCAM_IO_PBU		6	MDLX0		CAM_RST_N

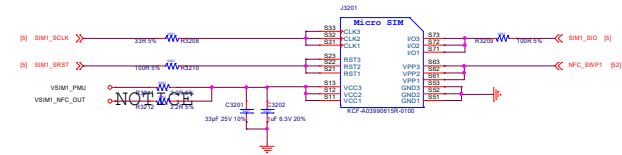
OVP



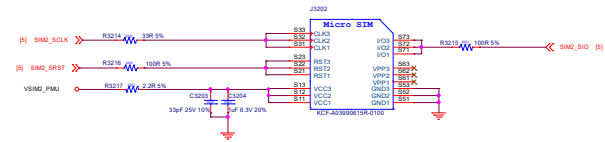
SD CARD



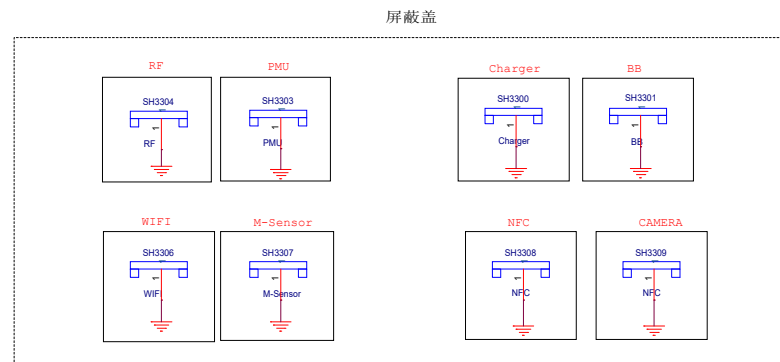
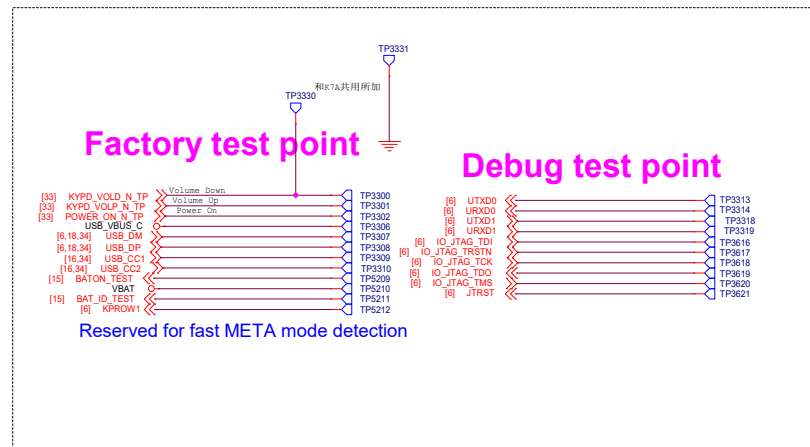
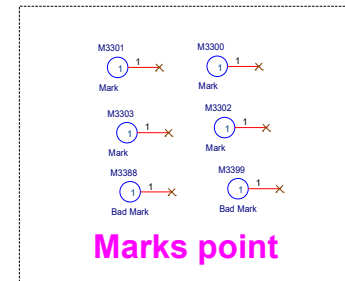
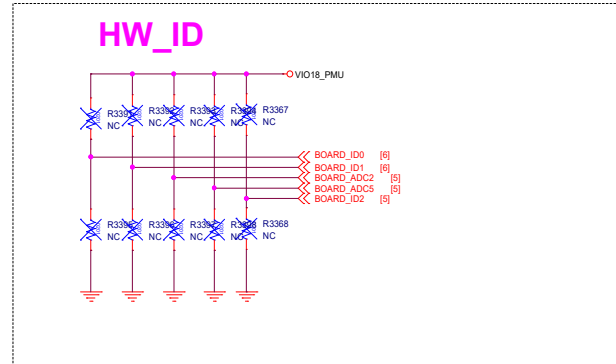
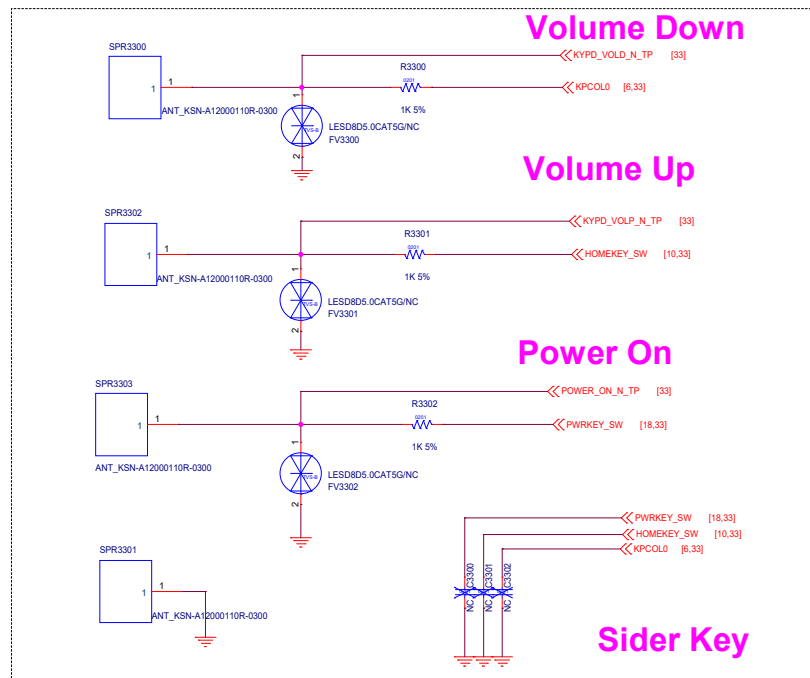
SIM1 PCB靠板边 SIM1

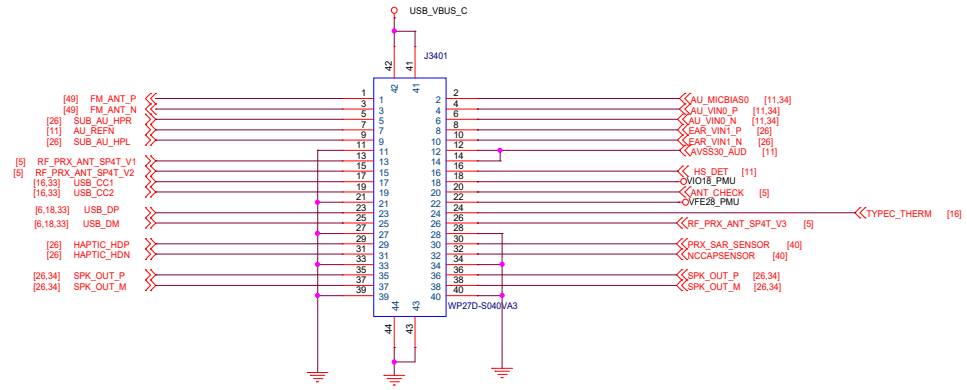


SIM2 PCB靠板内 SIM2

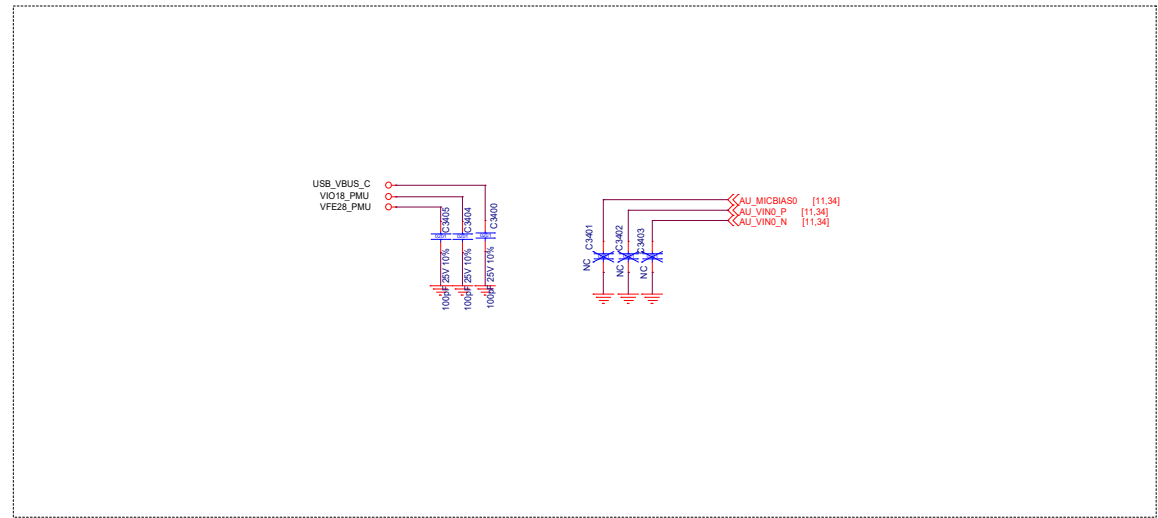


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Doc	Document Number
Doc	Doc
Date	Monday, November 14, 2020
Sheet	34 of 35



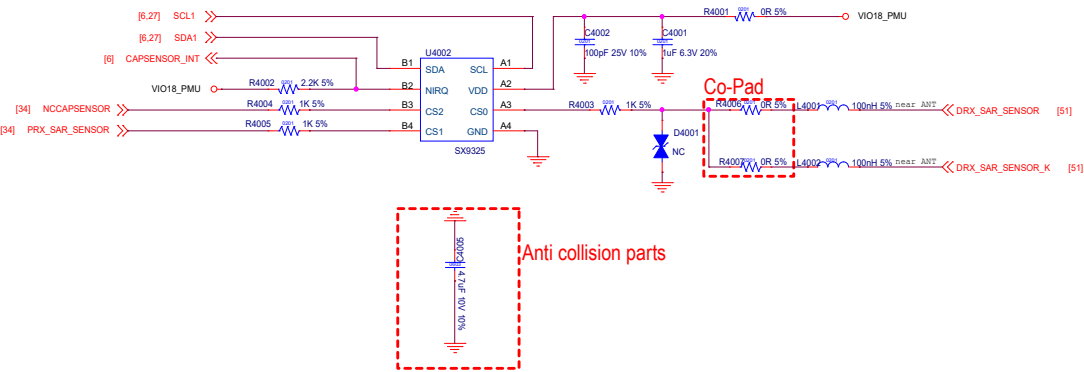


主副板连接器 1P1N位置还需要确定

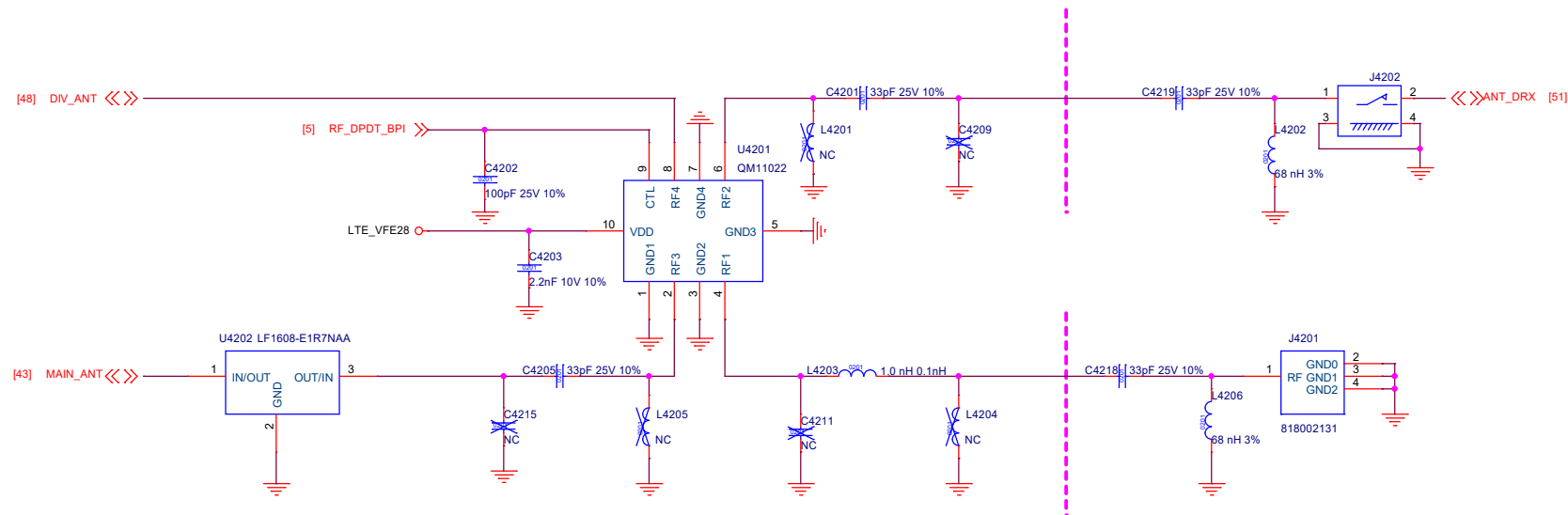


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Size C	Document Number <Doc>		Rev <RevCode>
Date:	Monday, December 14, 2020		Sheet 34 of 55

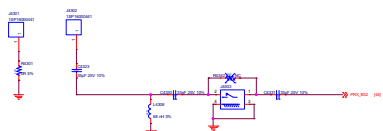
40_SAR_Sensor



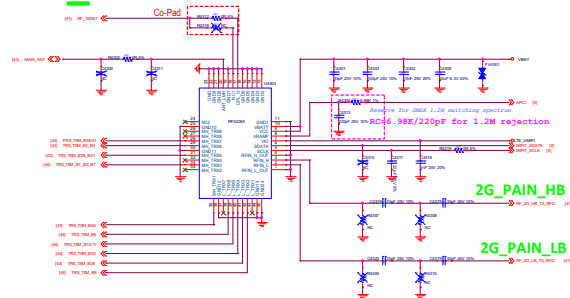
42_DPDT_Connector



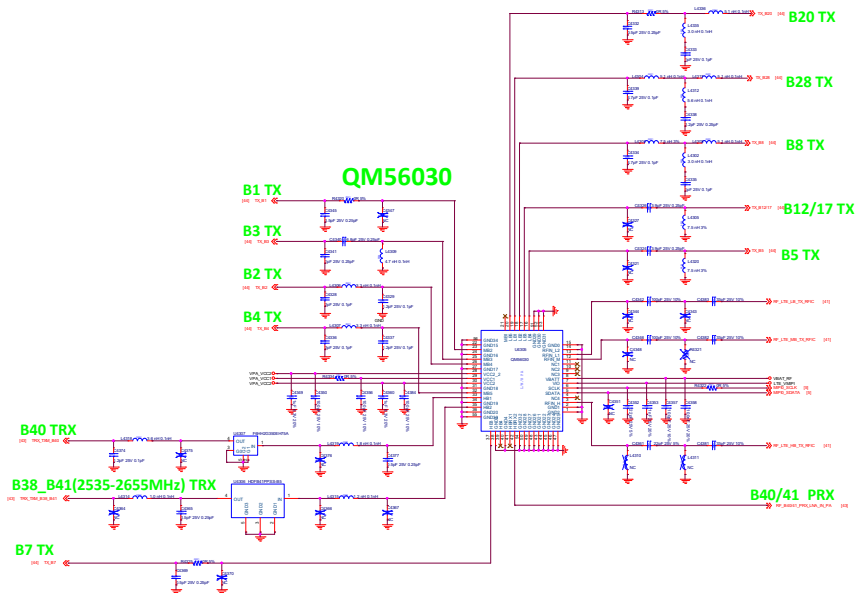
ANT_B32



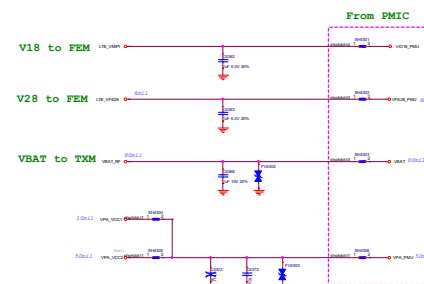
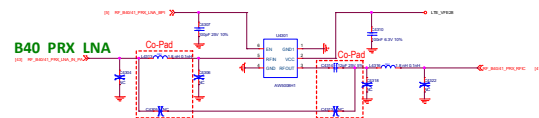
TXM_RF5228A



QM56030



B40 PRX LNA



Note:
Reserve one tuning cap. For PMIC VPA total cap requirement,
the total cap at RF side should be in 6.2uF +/-10%.

Rev	1.0
By	J. Kim
Check	Y. Kim
Date	2014.08.14

D

D

C

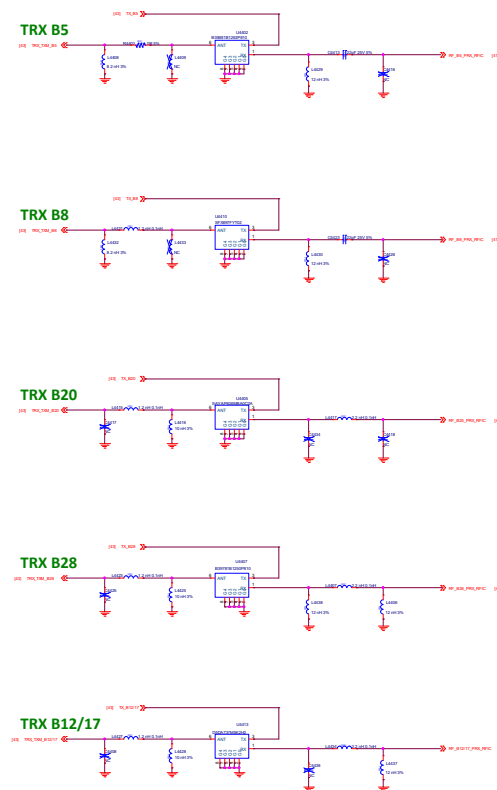
C

B

B

A

A



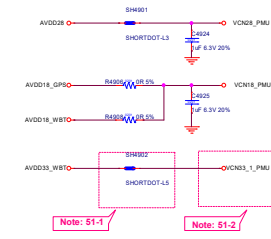
Title		AL_WF_SENSITIVE_WF_TX	
Doc ID	Document Number <Doc>		
Date	Monday, December 24, 2020	Time	02:47:00 AM

WIFI_2.4G

GPS

MT6631NA

Note: For MT6631 need to configure the VCN33_2_PMU to 2.8V by PMIC SW SW403

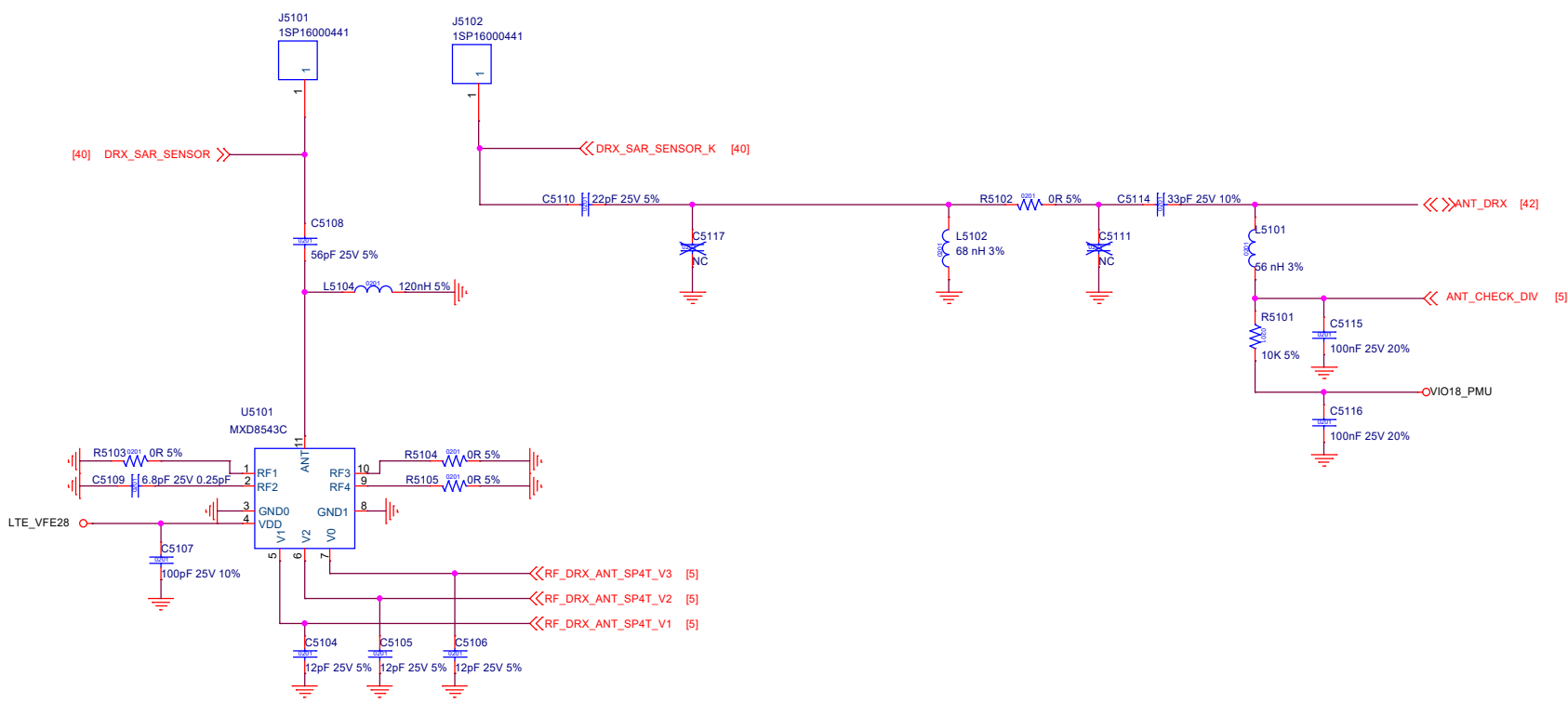


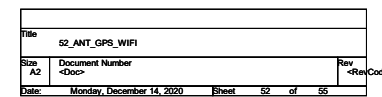
- Note 51-1: For R3712 size, please select 0402 size or larger one
- Note 51-2: Please refer to MT6765 Baseband design notice for VCN33 LDO selection guide
- Note 51-3: If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
- Note 51-4: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support

File	49_CONNECTIVITY_CONSYS_MT6631
Doc#	
Rev	Rev 1.0
Date	2023/11/14
Drawn	66
Check	67
Appr	68

Title			
50_ANT_NFC			
Size A3	Document Number <Doc>	Rev <Rev> Co	
Date:	Monday, December 14, 2020	Sheet	50 of 55

51_ANT_DIV





53_ANT_TOP_GND

